EcoCycle

A verified receipt for Lagos's waste economy.

Households are paid on the weight a facility scale actually recorded, not on anybody's word. Collectors get a track record they can prove. Processors get feedstock with **known provenance**. Every kilogram carries an audit trail from the doorstep to the payout.

736	15,225 kg	₦906,041	6,657
Batches tracked end to end	Verified on a facility scale	Settled to households and agents	Audited state changes, none untraced

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Submission · **August 2026** Built with · **PHP, Livewire, MySQL, OpenAI**

Status · **Working prototype, 90 days of history**

01 THE BUSINESS PROBLEM

Three people hold three different numbers, and none of them can be checked.

Lagos generates roughly **13,000 tonnes** of waste a day, and the recoverable part is already moving. Collectors are already working, households are already separating, processors are already buying. Nobody needs persuading that the material has value. What nobody has is a shared record of what moved, what it weighed, and who is owed for it.

The household is told a number

Bags are handed over and a weight is announced. There is no way to verify it, no way to know the rate, and no way to check the arithmetic afterwards.

The collector cannot prove anything

Paid on a figure declared at a gate, with no record of what was actually delivered. An honest collector has no way to demonstrate that they are honest.

The processor buys blind

Mixed loads of unknown provenance. A contaminated batch cannot be traced back to its source, so the cost is absorbed rather than corrected.

The gap becomes the business model

Because the three numbers are never reconciled, the spread between them is income for whoever is holding the scale.

The consequences compound. Households disengage permanently after being underpaid once. Collectors are treated as suspects rather than as a workforce. Processors cannot forecast feedstock, so they cannot invest in capacity. **The trust deficit, not the waste, is what caps the system.**

> **The constraint that shaped every decision:** one weight, held by one party, cannot do three incompatible jobs — planning the round, paying the collector, and paying the household. EcoCycle refuses to collapse them, and that single refusal produced the architecture on the next page.

02 THE WORKING PROTOTYPE

Three surfaces, one record, four rules nothing can work around.

Every figure on every screen is a live database query. There are no fixtures in the interface — this is not a mockup. The same record is read by a household on a phone, a collector on a round, and an inspector at a bench.

Household `/app`	Requests a pickup with **GPS-pinned coordinates**, follows the collector in live, sees the verified weight, holds a wallet balance and withdraws it.
Collection agent `/agent`	Ranked nearby jobs with real distances, a nearest-next route showing cumulative load, a five-stage pickup with QR tagging and photo capture, earnings and payouts.
Facility `/console`	Waste pipeline, inspection bench, batch registry, households and agents, settlements, analytics and integrity review.
Anyone `/b/{code}`	The public trace a QR sticker resolves to — **household, agent and inspector all read the identical record**.

Four rules, each enforced in exactly one place

Three weights `kept apart`	`expected_kg` (household predicts), `collected_kg` (agent declares), `verified_kg` (the scale). The gap between them is the fraud signal — collapsing them into one number throws that away.
Money settles `on verified only`	One service can value a batch. It refuses any batch that is not settleable and computes from the server's own rate table, never from a figure the browser sent. The household takes 60%; the agent takes the **remainder**, so rounding can never create or destroy a naira.
The ledger `is append-only`	There is no `wallet_balance` column anywhere. A balance is `SUM(credit) − SUM(debit)`, every write carries a unique idempotency key, and a mistake is corrected with a reversing entry rather than an update.
State changes `go through the machine`	Every transition is validated against the allowed next states, stamped, and written to an events table. Nothing else touches status — which is why **6,657 audited state changes** exist and not one batch is in an unknown state.

> **Honest about what is and is not AI.** Two features call a model. The other six "intelligence" modules are arithmetic and say so on screen: volume prediction is a weekday mean that reports its *measured* error, job matching ranks on distance and capacity, route ordering is nearest-next, and the integrity detectors are threshold rules. Labelling arithmetic as AI is the fastest way to lose a judge's trust.

03 THE SOLUTION WORKFLOW

From a bag on a doorstep to money in a wallet.

A batch is not a row that gets edited. It is an object that moves through a state machine and leaves an audit trail behind it. Every hand-off below is a validated transition.

1 ### The household requests a pickup

Picks a stream and an estimated weight, and pins the collection point from the device's GPS. Writes `expected_kg` only. The batch goes straight onto the marketplace.

2 ### An agent claims it

Jobs are ranked by haversine distance from the collector's live position and by remaining vehicle capacity. The claim guard sits in the SQL `WHERE` clause, so two agents racing for one job is settled by the database rather than by an optimistic interface.

3 ### The agent collects and declares

Five stages: travel, arrive, tag, weigh, done. Writes `collected_kg`. Variance against the household's estimate is computed as the agent types, so a large gap prompts a note at the doorstep rather than an argument at the gate an hour later.

4 ### The facility scans the tag and weighs the load

The batch arrives still marked in transit; the bench books it in and opens it in one motion. The inspector reads the scale and enters `verified_kg`. The settlement preview recomputes server-side as they type.

5 ### Money is released, once

Rate, value and both shares are frozen onto the batch, then two credits are written under keys derived from the batch id. Settling the same batch twice is a no-op. Four integrity detectors run on the same event.

6 ### Both parties withdraw

Unattached credits are locked, a payout row is created, and those credits are stamped with its id so they can never be paid twice. The facility exports a bank file and marks it paid.

> **The receipt is the product.** One event drives both an email and the in-app feed. The household sees the verified weight, the rate applied and their 60% share — the thing that did not exist before is a number they can check.

04 TOOLS, PLATFORMS AND AI MODELS

What it is made of.

AI models — two jobs, and everything else in code

JOB	MODEL	WHY THIS ONE
Analytics assistant	`gpt-4o-mini`	Only rewords figures already computed in SQL, so it needs no reasoning headroom. Cheap and fast is the right trade.
Photo classification	`gpt-4o-mini`	Estimates waste composition from a collection photograph, returned as strict JSON via the API's json_object mode.
Settlement	`no model`	Arithmetic against a server-side rate table. Free, instant and exactly right every time.
Integrity detection	`no model`	Four threshold rules — weight variance, GPS drift, volume spike, repeat rejection. Explainable to the agent they flag.

> **The model never touches the database and is never asked to compute.** Aggregates are calculated in SQL first and handed over as a fixed set of figures with an instruction to use nothing else. That is why an answer cannot invent a number — demonstrated on the next page.

Platforms and services

Laravel 13	Application framework, transactions, queues, notifications
Livewire 4	Server-rendered reactivity — settlement figures are never trusted from the browser
MySQL 9.1	Single system of record; row locking on payouts, atomic claim on jobs
OpenAI API	Analytics phrasing and photo classification
Resend	Transactional mail — the batch receipt, confirmed delivered
Geolocation API	Real device coordinates for pickup points and agent positions
Inline SVG QR	Batch tags render with no network round trip, so a collector with no signal can still tag a bag
PHPUnit	75 feature tests, plus three bespoke audit commands

Deliberately arithmetic, not machine learning

Haversine distance · nearest-next routing · weekday-mean forecasting that reports its own measured error · capacity-fit ranking · variance thresholds · GPS drift detection. Each is explainable to the collector it affects, which matters more at pilot scale than a marginal accuracy gain.

05 SAMPLE INPUTS AND OUTPUTS

Real exchanges, taken from the running system.

Everything below was produced by the live platform against its real database. Nothing is illustrative.

A. The assistant answering from figures it was given

INPUT · COMPUTED IN SQL, HANDED TO THE MODEL


```
{ "batch_count": 736, "verified_kg_total": 15225.6,
  "settled_month_naira": 906041.7, "reject_rate_pct": 6.1,
  "avg_variance_pct": -1.99, "by_category": { … } }
```


INPUT · THE OPERATOR'S QUESTION

What is our biggest waste stream by volume?

OUTPUT

"The biggest waste stream by volume is Organic waste, with a total weight of 8.37 tonnes."

generated **true** · source **from 736 batches · 12 areas**

B. The same assistant asked something its figures cannot support

INPUT

Which area is growing fastest, and can the plant absorb it?

OUTPUT · THE GUARDRAIL HOLDING

"There is insufficient information to determine which area is growing fastest, as growth rates or trends are not provided. Additionally, I cannot assess whether the plant can absorb the growth without data on the plant's maximum capacity."

Declined, and named exactly what was missing

> **This refusal is the most important output in the document.** Asked something its inputs cannot support, the model declines and says what is missing instead of producing a confident, plausible, invented figure. On a platform that pays people money, a system that would rather say nothing than guess is the design goal — not a limitation.

C. A batch through the inspection bench

INPUT · A LOAD ARRIVING OFF THE ROAD

```
batch     EC-W-2026-0092355
status    IN_TRANSIT
declared  10.2 kg        (what the agent says)
stream    Organic
```

OUTPUT · AFTER THE INSPECTOR WEIGHS IT

```
scale reading   9.3 kg
status          VERIFIED
value           NGN 1,097.40
  household 60% NGN   658.44
  agent     40% NGN   438.96

audit trail     IN_TRANSIT → RECEIVED → INSPECTING → VERIFIED
ledger entries  2, append-only
```

Shares reconstitute the batch value exactly

D. The state machine refusing illegal work

OUTPUT · GUARDS FIRING ON THE PIPELINE SCREEN

```
opened EC-W-2026-0092424 @ Requested
  moves offered                  Available, Cancelled
  audit row written              REQUESTED → AVAILABLE by admin

  illegal jump to Processed      refused
  verify without a scale reading refused
  reject without a reason        refused
  double settlement              no-op
  two agents race one job        database decides
```

06 EVIDENCE OF TESTING

What has actually been checked.

75	67	20	33
Feature tests, 240 assertions	Domain invariants, driven end to end	Invariants across every stored row	Pages rendered and floor-checked

Four layers, because each catches what the others cannot — all run against the real seeded MySQL database, not against mocks.

LAYER	COMMAND	WHAT IT PROVES
Feature suite	`artisan test`	Auth, role gates, four-step registration, and the full batch lifecycle driven through the real screens.
Domain invariants	`ecocycle:test`	Moves a real batch through inspection, settles it, tries to settle it twice, attempts an illegal transition, races two agents for one job. Runs inside a rolled-back transaction.
Stored-data audit	`ecocycle:audit`	Whether the rules hold across every row *already* stored, including rows written by code that has since changed. Read-only, safe against production.
Render audit	`ecocycle:pages`	Renders every page and asserts a plausible byte floor — a view that returns 400 bytes because a section quietly stopped rendering still returns 200.

The invariants that matter most

- The ledger's net **equals the sum of every party balance** — if these disagree, the append-only guarantee has broken somewhere.
- Household share plus agent share **reconstitutes the batch value** exactly, to the naira.
- **Nothing was valued without a verified weight**, and no rejected batch produced a ledger entry.
- Every batch carries an audit trail, and none is in an unknown state.

Verified interactively in a real browser

A render test cannot prove a button works, so the critical paths were driven through headless Chrome against the running app.

- Device geolocation captured and written to a batch — the pickup point moved from the registered address to the live device fix, **±9 m**.
- An agent position ping **re-ranked the jobs board**: the same three jobs went from 1.2 / 2.1 / 2.5 km to 3.3 / 9.6 / 20.3 km.
- Transactional mail confirmed **delivered** against the provider's API, rather than assumed from a lack of errors.

> **Testing found real defects, which is the point.** A settlement share misconfigured to 40/40 that the guard correctly refused to pay out on. A mail integration holding a valid API key under a name nothing read — silently logging instead of sending. A weighbridge preview that apportioned rounding to a different batch than the one that would actually receive it. A test suite that reports only passes is decoration.

07 DEMONSTRATION

Six minutes, one batch, three surfaces.

The demo is a single batch followed all the way through. Every screen shown is reading the same row — that is the whole argument. Password is `password` for every account.

SIGN IN AS	YOU LAND ON
Facility admin `manager@ecocycle.ng`	The full console, including settlement actions. Demo from this one.
Facility staff `inspector@ecocycle.ng`	The same console without settlement — proves the role gate is real.
Household `chidi.okonkwo@ecocycle.test`	A collection in flight: live map and the three weights filling in.
Collection agent `sunday.amadi@ecocycle.test`	Mid-round with a load on board; the pickup resumes where it was left.

The walkthrough

1. ### Open the public site

   Point out that the tonnage and settlement figures are live queries, not brochure copy.

2. ### As the household — request a pickup

   Press *Use my current location* and show the coordinates change. Submit, and show the QR batch tag.

3. ### As the agent — accept and collect

   The new job is on the board with a real distance. Accept it, then enter a declared weight *higher* than the household's estimate so the live variance appears.

4. ### As the facility — weigh it

   Take a scale reading *below* the declaration. Watch the settlement preview recompute server-side, then accept.

5. ### Show the flag it raised

   That variance has opened an integrity case against the agent, with the evidence attached.

6. ### Ask the analytics assistant two questions

   One it can answer, then one it cannot. **The refusal is the moment worth showing.**

7. ### Back as the household — withdraw

   The credit has landed at 60%. Withdraw it, then finish on the public trace URL: the same record, readable by anyone holding the bag.

> **The part worth watching** is the scale reading coming in below the declared weight — three numbers visibly disagreeing, money settling on the right one, and a flag opening automatically against the collector. That is the entire product in about fifteen seconds.

03 EXPECTED BUSINESS IMPACT

What changes for everyone in the chain.

₦543,625	₦362,416	6.1%	105
Credited to households on verified weight	Settled to collection agents	Rejection rate, measured not absorbed	Integrity cases raised with evidence

For the household	Turns an unpriced chore into **a receipt and an income**. Participation stops depending on trusting a collector's word, which is what makes separation at source durable rather than a one-off campaign.
For the collector	Converts informal work into **a verifiable track record**, each payment traceable to a batch. An honest collector can now prove it — the precondition for credit, insurance and better routes.
For the facility	Feedstock with **known provenance and known composition**. A contaminated load traces back to a household. Capacity planning runs on 15,225 kg of verified history instead of estimates.
For the operator	Fraud stops being an accepted cost. Mean declared-to-verified variance sits at **−1.99%**, so drift is visible per agent rather than lost in an aggregate.
For the city	Diversion becomes **auditable**. Because every kilogram is attributable end to end, recovery claims can be evidenced to a regulator or a carbon buyer.

What was deliberately not built

At pilot scale each of these is a line item with a known cost, not an oversight. **Weighbridge serial integration** — the inspector reads the scale and types it; integrity comes from the agent not being the one typing, not from a cable. **A payment gateway** — ledger, CSV export and mark-paid, until volume justifies the reconciliation work. **An offline agent app** — thirty-eight collectors; a dropped form is recoverable, five weeks of sync architecture is not yet warranted. **Four of eight integrity signals** — declared on the console and greyed out, because shipping detectors that never fire would look better and tell the operator less.

Honestly outstanding before a real roll-out: the classifier's agreement with inspectors has not been measured across a full intake day, and the rate table has not been validated against live offtake prices. Neither is a rebuild. Both are named here because a platform that pays people on a number it cannot defend is worse than no platform, and that is the risk worth being loudest about.

The thesis in one line. The material was always moving. EcoCycle gives the movement a record, and the record a value — then refuses to let anyone, including itself, write a number it cannot defend.